Lesion-Anchored Prototypes: Reducing Artifact Shortcuts in Interpretable Melanoma Classification

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***Abstract*—Melanoma classifiers often pick up acquisition artifacts such as dark corners, rulers and ink. In a prototype-based network the consequence is worse than a biased prediction: the explanation itself ("this part looks like that prototype") can point at the artifact. We first measure the problem on ISIC 2019: dark corners and bright specular structures are significantly associated with melanoma even after stratifying by acquisition source (Mantel–Haenszel odds ratios 1.56 and 1.75), and a black-box DenseNet-121 places its class-activation maps on artifacts 1.5 times more often than their area would predict. We then train a lesion-anchored prototypical part network with a lesion-attraction loss on prototype attention, an artifact-suppression loss and a projection restricted to lesion patches, using automatically generated masks instead of manual annotation. With an identical backbone, data split and training budget (three seeds), the method raises the test AUC of ProtoPNet from 0.825 to 0.865 (fine-annotation baseline: 0.844), projects every prototype onto lesion tissue, and lowers the share of its explanation hot region on artifacts from 6.6% to 4.8%. In controlled counterfactual tests that insert or remove artifacts while keeping every lesion pixel unchanged, it roughly halves the decision-flip rate of ProtoPNet for added dark corners and rulers (e.g., 10.9%→5.2%) and keeps its explanations stable (Spearman 0.92 vs. 0.68). According to our ablations, the gains come from lesion attraction and the constrained projection; the artifact-suppression term contributes little. Code, executed notebooks and logs are available at https://github.com/YousefAMovahed/lesion-anchored-prototypes.**

***Index Terms*—explainable AI, prototypical part networks, shortcut learning, dermoscopy artifacts, melanoma, counterfactual evaluation**

I. Introduction

Convolutional networks can classify skin lesions as accurately as dermatologists [1], but they also learn *shortcuts* [2], i.e. features that happen to correlate with the label in the training data without being related to the disease. Dermoscopy offers several documented cases. Surgical skin markings raise the melanoma probability of a deep classifier [3], and models still perform far above chance when the lesion is removed from the image [4], [5]. Dark corners, rulers, gel bubbles and ink are frequent in public archives and unevenly distributed across acquisition sites [6], [7].

For clinical use, Rudin [9] argues for models that are interpretable by design rather than explained after the fact. The prototypical part network (ProtoPNet) [10] compares image patches with learned prototypes, each projected onto a real training patch, so that a decision reads "this part looks like that part". ProtoPNet does not, however, stop a prototype from being a piece of a ruler or of a black corner. In that case the shortcut survives and is in addition presented to the user as a reasonable explanation [11]. Current fixes rely on expert annotation of the relevant regions [12] or on interactive feedback about single prototypes [13], and few studies test them by manipulating the artifacts directly.

Three questions guide this work. Are artifacts in ISIC 2019 associated with the diagnosis, and do standard and prototype models look at them? Can a prototype network be tied to lesion tissue without manual annotation? And do its decisions, and its explanations, stay the same when artifacts are added or removed while the lesion is left as it is? The paper makes four contributions:

- an audit of artifact bias on ISIC 2019 with automatically detected artifact masks, a leakage-free split and source-stratified statistics;
- a lesion-anchored ProtoPNet with lesion-attraction and artifact-suppression losses and a lesion-constrained projection, trained from automatic masks;
- a counterfactual protocol that inserts four artifact types and removes real ones with bit-identical lesion pixels and an inpainting-only null control, and measures both decision changes and explanation stability;
- a same-backbone comparison of eight configurations (three seeds each) on one internal and three external test sets, with ablations showing which components matter.

II. Related Work

Artifacts in dermoscopy. Bissoto *et al.* showed that ISIC models perform well even when the lesion is occluded [4], annotated seven artifact types [5], and used them to build domain-generalization benchmarks [6]. Winkler *et al.* [3] reported that skin markings change the output of a deep classifier. EPVT [7] learns artifact-domain prompts for a vision transformer, and Bevan and Atapour-Abarghouei [8] unlearn artifact information adversarially. All of these aim at accuracy under domain shift and leave open *where* an interpretable model looks.

Prototype networks and explanation supervision. ProtoPNet [10] learns class-specific latent prototypes, but the similar-

ity it suggests can be misleading [11]. IAIA-BL [12] penalizes prototype activation outside radiologist-marked regions, and ProtoPDebug [13] lets a user forbid confounded prototypes. Explanation penalties such as RRR [14] and CDEP [15] align gradient explanations with prior knowledge; CDEP was demonstrated on ISIC images with colored patches. In contrast, our method works without manual annotation, operates on the attention of the prototypes and on the projection step, and is evaluated by intervening on artifacts and observing both decisions and explanations; explanations, after all, have to be tested rather than taken at face value [16].

III. Method

A. Prototypical part network

We build on a DenseNet-121 backbone [17]. Two 1×1 convolutions with a final sigmoid turn its output into a latent map $z \in [0,1]^{H \times W \times D}$, here with $H = W = 7$ and $D = 128$. The network keeps $m = 20$ prototypes $p_j \in \mathbb{R}^D$, ten for each class, and $c(j)$ denotes the class of prototype j. For every cell we compute the distance $d_j(h,w) = \|z_{hw} - p_j\|_2^2$ and convert it into a similarity $s_j(h,w) = \log \frac{d_j(h,w)+1}{d_j(h,w)+\epsilon}$ with $\epsilon = 10^{-4}$. A prototype is as active as its best-matching cell, $a_j = \max_{h,w} s_j(h,w)$, and the logits are $\mathbf{W}\boldsymbol{a}$. Training minimizes

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_c \mathcal{L}_{\text{clst}} - \lambda_s \mathcal{L}_{\text{sep}} + \lambda_l \mathcal{L}_{\text{les}} + \lambda_a \mathcal{L}_{\text{art}}, \quad (1)$$

in which $\mathcal{L}_{\text{CE}}$ is a class-weighted cross-entropy, while the cluster and separation costs measure the distance to the closest prototype of the own and of the other class [10]. We keep the original weights $\lambda_c = 0.8$ and $\lambda_s = 0.08$ and set $\lambda_l = \lambda_a = 1$.

B. Automatic lesion and artifact masks

Lesion. HAM10000 comes with public lesion segmentation masks [18], which we use as they are for its 10,015 images. The remaining 15,316 images (BCN20000 [19] and MSK [20]) have no masks, so we predict them with a DeepLabV3 [21] model on a MobileNetV3-Large [22] encoder. This model saw only the HAM10000 masks of the training split. It reaches a Dice score of 0.945 on 1,852 held-out images and of 0.918 on PH2 [23], a set it never encountered.

Artifacts. We detect four artifact types at 256×256 pixels with simple deterministic operations; the dilated lesion is always excluded. *Dark corners* (DC) are dark regions that touch the image border and are at least 8 px thick on average. *Thin dark structures* (TS), which cover hair and ruler ticks, are found with a 13×13 black-hat filter, as in DullRazor [24], followed by an elongation test. *Bright structures* (SP), i.e. specular highlights and the rims of gel bubbles, come from an 11×11 white top-hat filter, and *ink or colored patches* (INK) from strongly saturated blue-to-violet pixels. We flag an image for a type once the mask covers more than 1%, 0.3%, 0.05% or 0.1% of it, respectively. When we inspected 24 randomly chosen flagged images per type, all 24 DC cases were correct and about 20 of 24 TS cases; the misses were mostly vessels and pores. The SP detector also fires on light hairs. Examples are shown in Fig. 1.

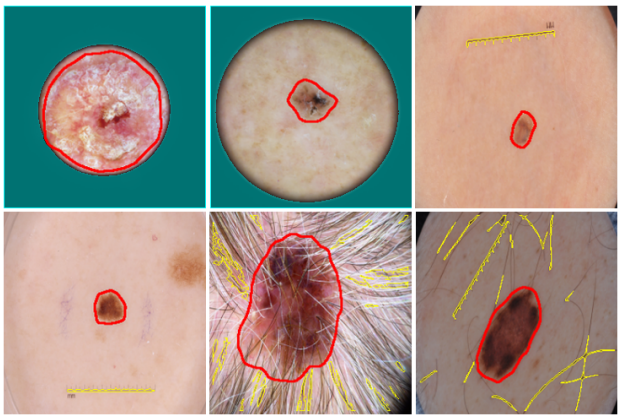


Fig. 1. Automatically generated masks on ISIC 2019 images (red: lesion). Top left and center: dark corners (shaded cyan); top right and bottom left: rulers and bottom center/right: hair, detected as thin dark structures (yellow). Artifact masks never overlap the lesion.

C. Lesion attraction and artifact suppression

During training the masks undergo the same geometric augmentation as the image. Average pooling to the 7×7 grid then gives, for each cell, the fraction covered by lesion, $M^L(h,w)$, and by artifacts, $M^A(h,w)$. We describe where a prototype looks by the spatial attention $\alpha_j = \text{softmax}_{h,w}(s_j/\tau)$, $\tau = 0.5$, and weight it by $\omega_j = a_j / \sum_k a_k$ (treated as a constant), so that the prototypes the model actually uses count most. With $A_{ij} = \sum_{h,w} \alpha_{ij} M_i^A$ and $L_{ij} = \sum_{h,w} \alpha_{ij} M_i^L$ for image i of a batch $\mathcal{B}$,

$$\mathcal{L}_{\text{art}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \sum_{j=1}^{m} \omega_{ij} A_{ij}, \quad (2)$$

$$\mathcal{L}_{\text{les}} = -\frac{1}{|\mathcal{B}_L|} \sum_{i \in \mathcal{B}_L} \sum_{j:\, c(j)=y_i} \tilde{\omega}_{ij} \log\left(L_{ij} + \delta\right), \quad (3)$$

where $\tilde{\omega}$ renormalizes ω over the prototypes of the true class y_i, $\delta = 10^{-4}$, and $\mathcal{B}_L$ holds the images whose crop still contains lesion. On its own, $\mathcal{L}_{\text{art}}$ could be minimized by shifting attention to healthy skin, which $\mathcal{L}_{\text{les}}$ rules out.

D. Lesion-constrained projection and last layer

In the "push" step, ProtoPNet replaces each prototype with the closest latent patch from a training image of the same class. We only allow cells with $M^L \geq 0.5$ and $M^A \leq 0.05$ as candidates. Every prototype is therefore a lesion patch without detected artifacts. After the push we train $\mathbf{W}$ alone and keep the backbone, the add-on layers and the batch-normalization statistics fixed, which keeps each prototype identical to the patch it came from. In our runs, five last-layer epochs were not enough to solve this convex problem for *any* of the prototype models. We therefore re-optimize $\mathbf{W}$ to convergence on fixed training activations (weighted cross-entropy plus an ℓ_1 penalty of 10^{-4} on negative-class connections [10]; 3,000 Adam steps). We apply this step in the same way to every prototype model, without touching the validation data, and also report results without it.

TABLE I
DATA (NUMBER OF IMAGES). THRESHOLDS AND CHECKPOINTS ARE SELECTED ON ISIC VALIDATION; ALL OTHER SETS ARE USED FOR TESTING ONLY.

Set	Benign	Melanoma	Total	Mel. %
ISIC 2019 train	16,895	3,657	20,552	17.8
ISIC 2019 validation	1,984	475	2,459	19.3
ISIC 2019 test	1,930	390	2,320	16.8
ISIC 2019 total, of which	20,809	4,522	25,331	17.9
source BCN20000	9,556	2,857	12,413	23.0
source HAM10000	8,902	1,113	10,015	11.1
source MSK	2,351	552	2,903	19.0
Derm7pt, dermoscopic	759	252	1,011	24.9
Derm7pt, clinical	759	252	1,011	24.9
PAD-UFES-20	2,246	52	2,298	2.3
PH2 (segmentation only)	–	–	200	–

Each source is split about 81/10/9% into train/validation/test.

IV. EXPERIMENTAL SETUP

A. Data and leakage-free split

The task is melanoma versus all other diagnoses on the 25,331 training images of ISIC 2019 (HAM10000 [18], BCN20000 [19] and MSK [20]), of which 4,522 (17.9%) are melanomas. All images are resized to 256×256. Unlike common practice, we keep the 2,084 MSK images that lack a lesion identifier. To avoid leakage we group images not only by lesion identifier but also by near-duplicate pairs (perceptual-hash distance ≤ 8 and a thumbnail correlation of at least 0.9 for both the full and the central image). Grouping look-alike pairs errs on the safe side, since it can only remove leakage. The 13,259 resulting groups are split into ten stratified folds, one for testing and one for validation (Table I); no group and no lesion identifier is shared between sets. As external test sets we use Derm7pt [25], which provides a dermoscopic and a clinical image for each of its 1,011 cases (melanomas include *in situ* lesions and metastases), and the smartphone images of PAD-UFES-20 [26]. PH2 serves only to check the segmentation, because the copy available to us has no diagnoses. None of the external images is used for training or for choosing thresholds or models.

B. Models and training

The eight configurations differ only in the components listed below; backbone (ImageNet-pretrained DenseNet-121), input size (224×224), data, split and schedule are the same. **CNN**: backbone, global average pooling and a linear layer (black box). **PPN-std**: ProtoPNet with standard augmentation (random crop, horizontal flip). **PPN**: ProtoPNet with strong augmentation (random resized crop, scale $[0.4, 1]$; horizontal and vertical flips; $\pm 20\%$ brightness, contrast, saturation), used by all following models. **PPN-FA**: PPN with the IAIA-BL fine-annotation loss [12] on our lesion masks ($\lambda = 10^{-3}$). **Ours**: Eq. (1) with constrained projection. **Ablations**: without $\mathcal{L}_{\text{les}}$ (Ours-noLes), without $\mathcal{L}_{\text{art}}$ (Ours-noArt), with standard projection (Ours-stdPush).

We train with Adam, a batch size of 64 and mixed precision. The first 5 epochs keep the backbone frozen and train the add-on layers and prototypes at a learning rate of 3×10^{-3}. The next 15 epochs train all layers (backbone 10^{-4}, weight decay 10^{-3}, learning rate divided by 10 every 5 epochs). The projection and 5 last-layer epochs follow; the CNN is trained for the same 20 epochs. Class weights are 0.608 and 2.81. For the CNN we keep the checkpoint with the best validation AUC, whereas prototype models are always evaluated after the projection. Each configuration is run with seeds 0, 1 and 2, which takes about 30 minutes per run on an NVIDIA T4.

C. Evaluation protocol

Classification. The threshold of each run is fixed on the ISIC validation set with the Youden index and then reused for every test set. AUC, balanced accuracy (BAcc), sensitivity and specificity are given as mean $\pm$ std over the seeds. For significance testing we average the probabilities of the three seeds (seed ensemble), compute 95% confidence intervals from 1,000 bootstrap resamples (resampling lesion groups on ISIC) and compare AUCs with the paired DeLong test [27], [28].

Attribution. For a prototype model we take the three most activated prototypes, upsample their similarity maps to 224×224 and keep the top 5% of pixels, i.e. the 95th-percentile region that ProtoPNet uses for visualization. For the CNN we use the class-activation map; with a global-average-pooling head it coincides with Grad-CAM [30] up to a positive factor [29]. From this hot region we compute the share that lies on the lesion and on artifacts, check whether its maximum falls on an artifact (pointing), and compute the *enrichment*, the artifact share divided by the artifact area. An enrichment above 1 means that artifacts get more attention than their size alone would explain.

Counterfactuals. Each test image is modified in four ways, always outside a 4-pixel margin around the lesion: we draw a ruler along one border, 12–30 curved hairs, a dark vignette, or 2–4 purple ink arcs. In the opposite direction, we erase the thin dark structures found by the detector (real hair and ruler marks, 544 images) with Telea inpainting [31]. As a null control, a region of the same shape is shifted onto artifact-free skin and inpainted as well. A check after every modification confirms that each lesion pixel is bit-identical to the original. We then record the mean $|\Delta p|$ of the melanoma probability and the rate of flipped decisions. For the three prototypes most active on the original image we also measure how much the explanation moves: the Spearman correlation ρ of the top-1 map, the IoU of the hot regions and the *capture*, the part of the new hot region that lies on inserted pixels. Seed-averaged per-image values are compared with the Wilcoxon signed-rank test.

V. RESULTS

A. Artifacts are associated with melanoma

In the training split, melanomas show dark corners and bright structures significantly more often than other lesions (Table II). This could simply reflect the source of the images, since dark corners appear in 47% of the BCN images but

TABLE II
ARTIFACT–MELANOMA ASSOCIATION (TRAINING SPLIT). OR: ODDS RATIO; MH: MANTEL–HAENSZEL OR STRATIFIED BY SOURCE.

Artifact	Prev.	$P(\mathrm{m}\|\mathrm{a})$	$P(\mathrm{m}\|\neg\mathrm{a})$	OR	MH OR [95% CI]
Dark corners	25.5%	.263	.149	2.03	1.56 [1.43, 1.69]
Bright structures	13.8%	.266	.164	1.85	1.75 [1.59, 1.92]
Thin dark struct.	24.2%	.174	.179	0.97	0.99 [0.90, 1.08]
Ink / patches	1.6%	.202	.178	1.18	1.11 [0.85, 1.46]
Any	48.8%	.210	.148	1.53	1.27 [1.17, 1.36]

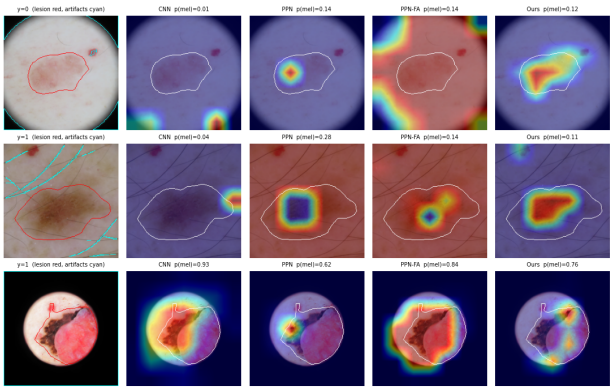


Fig. 2. ISIC test images with artifacts (left: lesion red, detected artifacts cyan). CNN: class-activation map; prototype models: map of the most activated prototype; p: melanoma probability (seed-0 models). Here the CNN and PPN-FA respond to dark corners and borders, whereas the top prototype of Ours stays on the lesion.

in only 5% of the HAM images, and BCN also has the highest melanoma rate. Stratifying by source does not remove the effect, however. The Mantel–Haenszel odds ratios remain clearly above 1, and for dark corners the odds ratio exceeds 1 in each source separately (BCN 1.41, HAM 3.10, MSK 2.24). For thin dark structures and ink we find no association.

B. Classification

On the ISIC test set, Ours reaches the highest AUC among the prototype models (0.865 ± 0.006, ensemble 0.884; Table III). The difference to PPN, PPN-std and PPN-FA is significant (DeLong $p \leq 7 \times 10^{-4}$), and the results vary less between seeds (std 0.006 compared with 0.040 for PPN). The black-box CNN is still more accurate (ensemble 0.912), so interpretability keeps a price of about 3 AUC points; for PPN the gap is 4.6. The ranking does not depend on the last-layer re-optimization; with the original five-epoch last layer the test AUCs are 0.846 (Ours), 0.770 (PPN), 0.765 (PPN-FA) and 0.797 (PPN-std). Externally, Ours has the highest prototype-model AUC on Derm7pt-dermoscopic (0.749; ensemble $p = 1.7 \times 10^{-5}$ vs. PPN-FA, 2.8×10^{-6} vs. PPN-std, 0.20 vs. PPN, whose seed variance is high) and Derm7pt-clinical (0.657; $p = 0.046$ vs. PPN). PAD-UFES-20 contains only 52 melanomas, and there none of the prototype models differs significantly from Ours. Excluding the external images flagged as possible ISIC near-duplicates (8, 6 and 7) changes each external AUC by at most 0.004.

C. Where the models look

The black-box CNN is the model most drawn to artifacts (Table IV, Fig. 2). In images that contain artifacts, 23.1% of its hot region lies on them (36.5% when dark corners are present), 1.5 times more than their area would suggest, and in 11.7% of the test images its maximum sits on an artifact. The prototype models look at artifacts less than their area would suggest, and Ours does so least: 4.8% of the hot region (PPN 6.6%, PPN-FA 10.2%), an artifact maximum in 1.3% of images (PPN 4.2%), and the largest lesion fraction (58.9%). All 20 prototypes of every Ours run are lesion patches, whereas PPN-FA projects on average 10.0 prototypes onto lesion and 2.0 onto artifact cells (Fig. 3). On Derm7pt-dermoscopic the artifact pointing rate is again lowest for Ours (1.5% vs. 4.4% for PPN and 15.3% for the CNN), although PPN places a larger hot-region fraction on the lesion there (0.79 vs. 0.73).

D. Counterfactual robustness

Table V summarizes the interventions; in all 2,063–2,318 modified images per insertion type the lesion pixels were confirmed to be unchanged. Adding dark corners *increases* the melanoma probability of every model (mean signed change $+0.041$ CNN, $+0.048$ PPN, $+0.020$ Ours), which matches the sign of the association in Table II. The models have absorbed the shortcut, and Ours has absorbed it least. Ours roughly halves the flip rate of PPN for dark corners (10.9%→5.2%) and rulers (7.5%→3.6%), reduces it for ink (9.5%→5.6%) and hair (14.0%→9.2%), and flips fewer decisions than the CNN for all four insertions. All per-image $|\Delta p|$ differences between Ours and PPN are significant (Wilcoxon $p < 10^{-39}$), as are those against the CNN ($p \leq 0.018$). Removing real hair and ruler marks changes more decisions than the null control for every model (Ours 4.4% vs. 1.6%), so the measured sensitivity stems from the artifacts, not from inpainting.

Its explanations also move the least (Fig. 4). After adding dark corners its top-1 prototype map keeps $\rho = 0.92$ (PPN 0.68), and only 3.4% of its hot region moves onto the inserted pixels (PPN 6.1%, PPN-FA 11.4%; $p < 10^{-100}$). The exception is ink: the arcs lie just outside the lesion border, where lesion-anchored prototypes respond, so 1.8% of the hot region of Ours falls on ink vs. 1.0% for PPN, although Ours still flips fewer decisions.

E. Ablation

Lesion attraction turns out to be the decisive term. Without it (Ours-noLes) the test AUC falls to 0.830 (DeLong $p = 8 \times 10^{-8}$), the artifact share of the hot region more than doubles (4.8%→10.6%), and the flip rates for dark corners, rulers and ink roughly double. Removing artifact suppression (Ours-noArt) degrades no metric: the AUC difference is not significant ($p = 0.085$), attribution is equivalent, and robustness is similar or slightly better (e.g., $|\Delta p|$ 0.061 vs. 0.081 for hair). With attention anchored to the lesion and the projection constrained, $\mathcal{L}_{\mathrm{art}}$ brings no measurable benefit, so it can be left out. The constrained projection does not change accuracy (Ours-stdPush, $p = 0.85$) but determines what prototypes *are*:

TABLE III

CLASSIFICATION (MEAN ± STD OVER 3 SEEDS; THRESHOLD FROM ISIC VALIDATION). ENS.: SEED-ENSEMBLE AUC WITH 95% BOOTSTRAP CI; p: PAIRED DELONG TEST OF THE ENSEMBLE AGAINST OURS ON ISIC TEST. BOLD: BEST PROTOTYPE MODEL.

	ISIC 2019 test (n = 2,320)						External AUC		
Model	AUC	BAcc	Sens.	Spec.	Ens. AUC [95% CI]	p	Derm7pt-derm	Derm7pt-clin	PAD-UFES-20
CNN (black box)	.902±.006	.814±.011	.808±.044	.820±.040	.912 [.884, .935]	3e-5	.799±.023	.704±.022	.721±.022
PPN-std	.835±.013	.752±.014	.769±.063	.734±.040	.861 [.827, .888]	6e-6	.677±.027	.624±.023	.587±.019
PPN	.825±.040	.758±.010	.786±.068	.730±.087	.866 [.830, .894]	7e-4	.688±.072	.618±.038	.665±.016
PPN-FA (IAIA-BL)	.844±.008	.744±.021	.689±.045	.800±.005	.866 [.830, .895]	4e-4	.696±.024	.640±.013	.630±.059
Ours	**.865±.006**	**.780±.014**	.734±.075	**.825±.046**	**.884 [.850, .910]**	–	**.749±.009**	**.657±.019**	.647±.108
Ours-noLes	.830±.015	.746±.030	.766±.091	.726±.032	.856 [.821, .885]	8e-8	.717±.016	.636±.033	.628±.047
Ours-noArt	.859±.011	.772±.003	.733±.056	.811±.054	.877 [.845, .904]	.085	.752±.014	.637±.030	.678±.018
Ours-stdPush	.863±.005	.785±.002	.823±.043	.747±.047	.884 [.852, .911]	.85	.747±.005	.670±.019	.658±.108

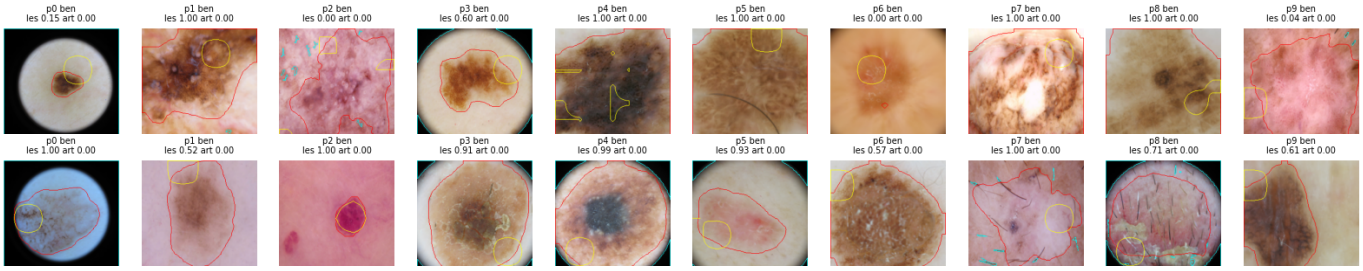


Fig. 3. Source patches of the ten benign-class prototypes of PPN (top) and Ours (bottom), seed 0 ("les"/"art": lesion and artifact fraction of the projected cell; red: lesion; cyan: artifacts; yellow: 95th-percentile region of the prototype on its own source image). Four PPN prototypes come from background skin (lesion fraction 0.00–0.15); all prototypes of Ours lie on lesion tissue.

TABLE IV

WHERE THE MODELS LOOK (ISIC TEST, MEAN ± STD OVER SEEDS). ART.: HOT-REGION FRACTION ON ARTIFACTS IN IMAGES WITH ARTIFACTS; ENR.: ARTIFACT ENRICHMENT; POINT: MAXIMUM ON AN ARTIFACT; L-PR.: PROTOTYPES PROJECTED ONTO LESION CELLS (OF 20).

Model	Lesion↑	Art.↓	Enr.↓	Point↓	L-pr.
CNN	.466±.040	.231±.024	1.52±.46	.117±.007	–
PPN-std	.518±.036	.082±.049	0.54±.31	.050±.029	12.0
PPN	.562±.012	.066±.045	0.46±.04	.042±.040	14.3
PPN-FA	.530±.051	.102±.024	0.68±.05	.050±.006	10.0
Ours	**.589±.040**	**.048±.015**	**0.37±.13**	**.013±.005**	**20.0**
Ours-noLes	.565±.051	.106±.056	0.67±.11	.053±.029	20.0
Ours-noArt	.594±.028	.042±.008	0.40±.03	.018±.007	20.0
Ours-stdPush	.515±.063	.086±.052	0.58±.30	.034±.024	7.7

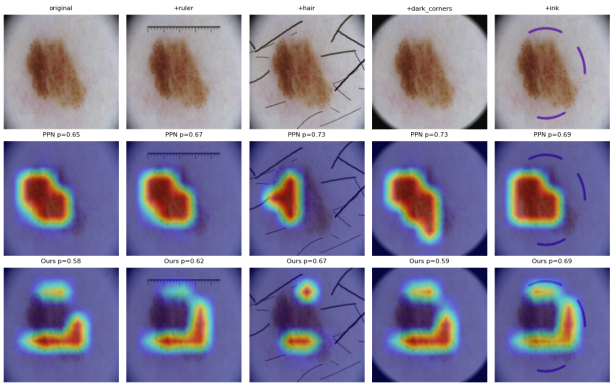


Fig. 4. Counterfactual example with unchanged lesion pixels. Rows 2–3: map of the prototype most active on the original image and melanoma probability p for PPN and Ours.

with the standard projection only 7.7 of 20 prototypes lie on the lesion, the artifact share rises from 4.8% to 8.6%, and the capture by added dark corners from 3.4% to 7.5% ($p < 10^{-200}$).

VI. DISCUSSION AND LIMITATIONS

Tying prototype attention to the lesion made the prototype network both more accurate and more faithful, and restricting the projection ensures that its explanations are built from lesion tissue. The study has several limitations: (i) a gap of about 3 AUC points to the black-box CNN remains; (ii) masks are automatic: predicted lesion masks of BCN20000 and MSK are not checked against expert masks, the bright-structure detector also responds to light hairs, and other artifacts (e.g., gel borders) are not modeled; (iii) inserted artifacts are synthetic, which we mitigated with real-artifact removal and a null control; (iv) the 7×7 latent grid limits spatial resolution; (v) external gains are significant on Derm7pt but not on the small PAD-UFES-20 melanoma set, and a single backbone was studied, as required for a controlled comparison.

VII. CONCLUSION

Standard and prototype classifiers trained on ISIC 2019 do exploit artifact shortcuts, and a prototype network can be anchored to the lesion without manual annotation. In our same-backbone comparison the lesion-anchored model was the most

TABLE V

COUNTERFACTUAL INTERVENTIONS ON ISIC TEST (LESION PIXELS UNCHANGED; MEAN OVER 3 SEEDS). FLIP: DECISION CHANGES (%); $|\Delta p|$: MEAN ABSOLUTE CHANGE OF THE MELANOMA PROBABILITY; ρ: SPEARMAN CORRELATION OF THE TOP-1 PROTOTYPE MAP BEFORE/AFTER; CAP.: SHARE OF THE NEW HOT REGION ON INSERTED PIXELS (%).

	+Dark corners		+Ruler		+Hair		+Ink		Remove hair/ruler		Null control		Expl.: dark corners		Expl.: ruler	
Model	Flip	$\lvert\Delta p\rvert$	Flip	$\lvert\Delta p\rvert$	Flip	$\lvert\Delta p\rvert$	Flip	$\lvert\Delta p\rvert$	Flip	$\lvert\Delta p\rvert$	Flip	$\lvert\Delta p\rvert$	$\rho\uparrow$	Cap.$\downarrow$	$\rho\uparrow$	Cap.$\downarrow$
CNN	6.1	.049	5.6	.049	12.5	.104	9.8	.080	4.7	.036	1.0	.008	–	–	–	–
PPN-std	10.9	.060	8.3	.041	14.9	.092	9.9	.058	8.8	.036	4.3	.013	.635	7.1	.773	1.5
PPN	10.9	.067	7.5	.045	14.0	.096	9.5	.062	7.4	.045	3.6	.019	.680	6.1	.804	1.1
PPN-FA	10.6	.049	6.5	.028	15.1	.084	9.1	.046	7.2	.030	3.2	.010	.714	11.4	.874	1.1
Ours	**5.2**	**.035**	**3.6**	**.025**	**9.2**	**.081**	**5.6**	**.045**	**4.4**	.032	1.6	.011	**.918**	**3.4**	**.952**	**0.7**
Ours-noLes	10.2	.053	8.7	.040	14.3	.083	10.1	.049	8.3	.042	3.9	.016	.705	8.6	.827	1.7
Ours-noArt	5.4	.036	3.5	.023	8.7	.061	6.1	.041	4.4	.027	1.0	.011	.884	3.4	.949	1.0
Ours-stdPush	4.5	.034	4.1	.028	9.6	.077	5.8	.045	4.5	.033	1.9	.011	.923	7.5	.944	1.4

accurate prototype model, relied least on artifacts, and changed its decisions and explanations least when artifacts were inserted or removed around an unchanged lesion. Because the counterfactual protocol does not depend on the architecture, it can also be used to check the faithfulness of other interpretable classifiers.

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